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Resume_Screening_System.ipynb

Support_Ticket_Classification.ipynb

C: > Users > divya > Downloads > Resume_Screening_System.ipynb > import pandas as pd

GenerateCodeMarkdownRun AllClear All OutputsOutlineSelect Kernel

df.head()

[21]Python

(9544, 35)

	address	career_objective	skills	educational_institution_name	degree_names	passing_years	educational_results	result_types	major_field_of_studies	professional_company_names
0	NaN	Big data analytics working and database wareho...	['Big Data', 'Hadoop', 'Hive', 'Python', 'Mapr...']	['The Amity School of Engineering & Technology...']	['B.Tech']	['2019']	['N/A']	[None]	['Electronics']	['Coca-COLA']
1	NaN	Fresher looking to join as a data analyst and ...	['Data Analysis', 'Data Analytics', 'Business ...']	['Delhi University - Hansraj College', 'Delhi ...']	['B.Sc (Maths)', 'M.Sc (Science) (Statistics)']	['2015', '2018']	['N/A', 'N/A']	['N/A', 'N/A']	['Mathematics', 'Statistics']	['BIB Consultancy']
2	NaN	NaN	['Software Development', 'Machine Learning', '...']	['Birla Institute of Technology (BIT), Ranchi']	['B.Tech']	['2018']	['N/A']	['N/A']	['Electronics/Telecommunication']	['Axis Bank Limited']
3	NaN	To obtain a position in a fast-paced business ...	['accounts payables', 'accounts receivables', ...]	['Martinez Adult Education, Business Training ...']	['Computer Applications Specialist Certificate...']	['2008']	[None]	[None]	['Computer Applications']	['Company Name i¼ City , State', 'Company Name...']
4	NaN	Professional accountant with an outstanding wo...	['Analytical reasoning', 'Compliance testing k...']	['Kent State University']	['Bachelor of Business Administration']	[None]	['3.84']	[None]	['Accounting']	['Company Name', 'Company Name', 'Company Name...']

5 rows x 35 columns

Spaces: 4

LF

{ }

Cell 1 of 15

Go Live

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[24]

Python

▷ ▽

```
df['clean_resume'] = (
    df['skills'].astype(str)
    + " " +
    df['career_objective'].astype(str)
)

df['clean_resume'] = df['clean_resume'].apply(clean_text)

df[['clean_resume']].head()
```

[25]

Python

clean_resume

- 0 big data hadoop hive python mapreduce spark ja...
- 1 data analysis data analytics business analysis...
- 2 software development machine learning deep lea...
- 3 accounts payables accounts receivables account...
- 4 analytical reasoning compliance testing knowle...

```
def extract_skills(text):  
  
    skills = str(text).lower().split(',')  
  
    skills = [s.strip() for s in skills if s.strip()]  
  
    return set(skills)
```

[26]

Python

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website.HTML

Resume_Screening_System.ipynb

Support_Ticket_Classification.ipynb

C: > Users > divya > Downloads > Resume_Screening_System.ipynb > df['resume_skill_set'] = df['skills'].apply(extract_skills)

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Select Kernel

df['resume_skill_set'] = df['skills'].apply(extract_skills)

df['required_skill_set'] = df['skills_required'].apply(extract_skills)

df[['resume_skill_set', 'required_skill_set']].head()

[27]Python

	resume_skill_set	required_skill_set
0	{'data structures', 'python', 'yarn', 'informa...	{}
1	{'machine learning', 'tableau', 'sas', 'data ...	{}
2	{'requirement gathering', 'html', 'c', ['softw...	{'brand promotion\ncampaign management\nfield s...
3	{'powerpoint', 'insurance', ['accounts payable...	{'fast typing skill\nieltsinternet browsing & o...
4	{'research', 'client', 'policies', 'clients', ...	{'ios\nios app developer\nios application devel...

df['matched_skills'] = df.apply(
 lambda row:
 list(
 row['resume_skill_set'].intersection(
 row['required_skill_set']
)
),
 axis=1
)

df[['matched_skills']].head()

[28]Python

matched_skills

0

0 13

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```
df['matched_skills'] = df.apply(
    lambda row:
        list(
            row['resume_skill_set'].intersection(
                row['required_skill_set']
            )
        ),
    axis=1
)
```

```
df[['matched_skills']].head()
```

[28]

Python

...	matched_skills
0	☐
1	☐
2	☐
3	☐
4	☐

▷

```
df['missing_skills'] = df.apply(
    lambda row:
        list(
            row['required_skill_set']
            -
            row['resume_skill_set']
        ),
    axis=1
```

[29]

Python

2	U
3	[]
4	[]

```
df['missing_skills'] = df.apply(
    lambda row:
        list(
            row['required_skill_set']
            -
            row['resume_skill_set']
        ),
    axis=1
)

df[['missing_skills']].head()
```

[12] Python

	missing_skills
0	[]
1	[]
2	[brand promotion\ncampaign management\nfield s...
3	[fast typing skill\nieltsinternet browsing & o...
4	[ios\nios app developer\nios application devel...

```
df['skill_match_score'] = df.apply(
    lambda row:
        (
            len(row['matched_skills'])
            /

```

[13] Python

```
def extract_skills(row):  
    /  
    len(row['required_skill_set'])  
    * 100  
    )  
    if len(row['required_skill_set']) > 0  
    else 0,  
    axis=1  
)  
  
df[['skill_match_score']].head()
```

[30] Python

...

	skill_match_score
0	0.0
1	0.0
2	0.0
3	0.0
4	0.0

```
ranking = df.sort_values(  
    by='skill_match_score',  
    ascending=False  
)  
  
ranking[  
    'skill_match_score',  
    'matched_skills',  
    'missing_skills'  
]
```

[31] Python

Generate + Code + Markdown Run All Clear All Outputs Outline ...

 Select Kernel

```
    'matched_skills',  
    'missing_skills'  
]  
].head(10)
```

[31]

Python

...	skill_match_score	matched_skills	missing_skills
9543	0.0	[]	[autocad\netabs\nmicrosoft office suite\nms pr...
0	0.0	[]	[]
1	0.0	[]	[]
2	0.0	[]	[brand promotion\ncampaign management\nfield s...
3	0.0	[]	[fast typing skill\nielsinternet browsing & o...
4	0.0	[]	[ios\nios app developer\nios application devel...
5	0.0	[]	[python\nr or java\ntensorflow\npytorch\nsciki...
6	0.0	[]	[ios\nios app developer\nios application devel...
9527	0.0	[]	[azure\nbig data\ndata analytics\netl tools\np...
9526	0.0	[]	[autocad\netabs\nmicrosoft office suite\nms pr...

```
tfidf = TfidfVectorizer(max_features=3000)

X = tfidf.fit_transform(
    df['clean_resume']
)

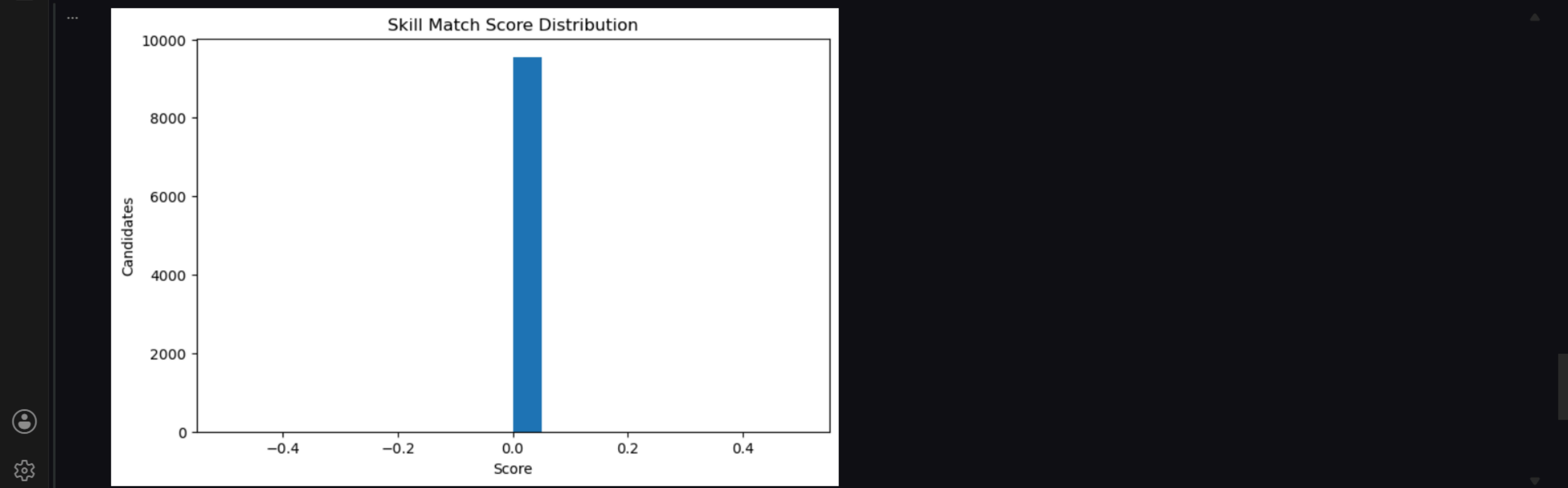
print(X.shape)
```

[32]

Python

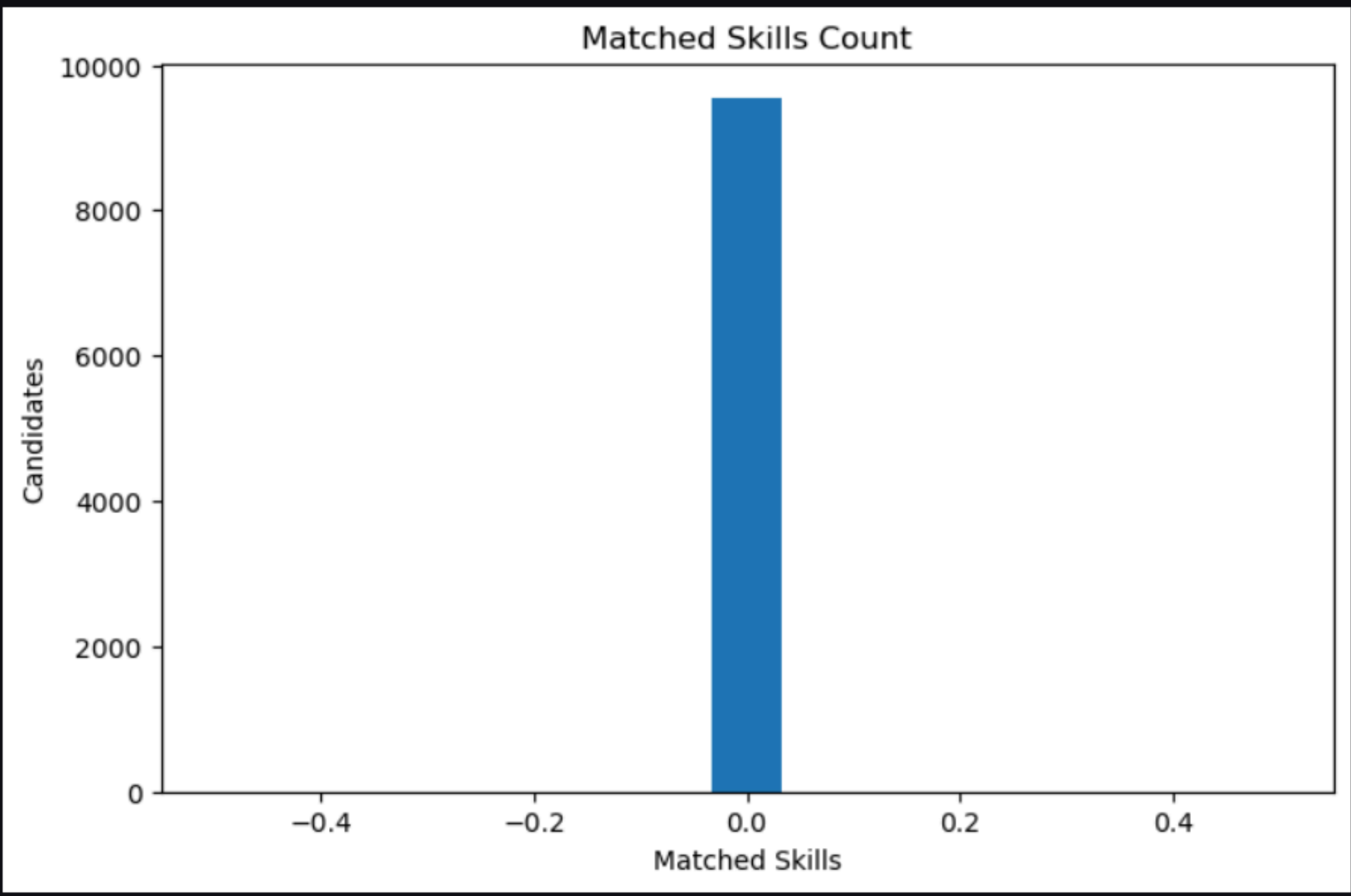
... (9544, 2955)

```
),  
plt.title("Skill Match Score Distribution")  
  
plt.xlabel("Score")  
  
plt.ylabel("Candidates")  
  
plt.show()
```



```
plt.ylabel("Candidates")  
  
plt.show()
```

[17] Python



▷ ∨

[19]

Breaking

0.0000

	skill_match_score	matched_skills	missing_skills
9543	0.0	[]	[autocad\netabs\nmicrosoft office suite\nms pr...
0	0.0	[]	[]
1	0.0	[]	[]
2	0.0	[]	[brand promotion\ncampaign management\nfield s...
3	0.0	[]	[fast typing skill\nielsinternet browsing & o...
4	0.0	[]	[ios\nios app developer\nios application devel...
5	0.0	[]	[python\nr or java\ntensorflow\npytorch\nsciki...
6	0.0	[]	[ios\nios app developer\nios application devel...
9527	0.0	[]	[azure\nbig data\data analytics\netl tools\np...
9526	0.0	[]	[autocad\netabs\nmicrosoft office suite\nms pr...
9525	0.0	[]	[having cacc from reputed ca firm\ninternal au...
9524	0.0	[]	[python\nr or java\ntensorflow\npytorch\nsciki...
9523	0.0	[]	[audit and inspection\nbanking\ninternal audit]
9522	0.0	[]	[brand promotion\ncampaign management\nfield s...
9521	0.0	[]	[fast typing skill\nielsinternet browsing & o...
9520	0.0	[]	[]
9519	0.0	[]	[autocad\nsolidworks]
9518	0.0	[]	[audit and inspection\nbanking\ninternal audit]
9517	0.0	[]	[human resource management]
9516	0.0	[]	[]